3D Vision - Lab 3

Student Names		
Student IDs		
Group		

*Note: **Resizing answer boxes will be penalized**. Ensure your response stays within the provided borders to ensure compatibility with Gradescope.*

The geometry of two views and 3D reconstruction

1. Estimation of the fundamental matrix with the 8-point algorithm

Paste the code for normalization function	• Explain the normalization process. • Which kind of transformations is this homography applying? • Why do we need normalization?

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Paste the code for 8-point algorithm function	• Explain the algorithm step by step. • Why do we need to enforce the rank 2 constraint? • How do we enforce it? • What would happen if rank(F) is 3?

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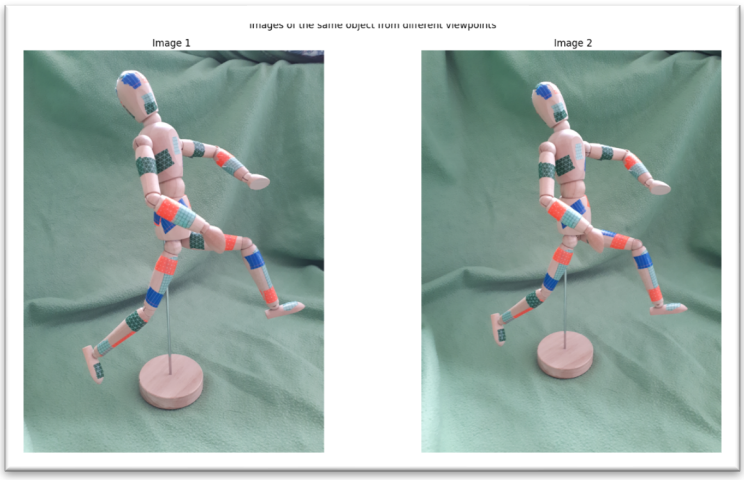
2. Robust estimation of the Fundamental matrix



Paste the obtained SIFT correspondences
• How were these correspondences obtained? • Are there any incorrect correspondences? Spot them.

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• Paste the code for compute_inliers() function.
• What is the distance computed to estimate the error? Why is it not the Euclidean distance? • When will the numerator of the distance have small values? And big ones? • How do you select the inliers?

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Paste the figure showing the inliers.
• Briefly explain how RANSAC works and why we use it. • How many iterations do you need to find F? Why? • How many inliers have you found? Are they properly identified? Why?

Epipolar lines and epipoles

- How did you obtain the epipolar lines?
- Where do they intersect?

- How do you obtain the epipoles from the fundamental matrix? Why?
- Compare the epipoles obtained with the intersection of the epipolar lines. Paste their values. Are they similar? Should they be similar?
- Compare the lines obtained by joining the epipole with the matched point and the previous corresponding epipolar lines. Are they similar? Why? Should they be similar?

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2. Structure computation

2.1. Triangulation

Explain the theoretical process to obtain the A matrix, the system of equations to solve triangulation.
Paste the code completed for triangulation.

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2.2 Relative motion between the two cameras

- Write the formula to obtain the Essential matrix
- What does the essential matrix represent?
- Paste the obtained Essential matrix

- Write the first camera projection matrix P1. Why have we defined it like that?

Paste the plot of potential camera projection matrices for the second camera (P2)

- Why do we have 4 possible P2 matrices?
- How did you obtain them?

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Paste the plot for “Camera 1 and Selected Camera 2”
• How did we choose the right solution from the four candidates? • The code to do it is provided, explain how it works. • Looking at the original pictures, do the estimated camera positions make sense to you?

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2.3. Reconstruction from 2 views

Paste different viewpoints of the 3D reconstruction
• Does it look right to you? Why? • Can you spot points in the manikin, the base of the manikin, and in the green background chroma? • Can you spot any incorrect point?

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Paste the reprojection error code
Paste the reprojection error histogram
• Interpret the results of the reprojection error histogram above • What does it mean? • Do you obtain reasonable error values?

3. 3D-Reconstruction of your own pair of images

Paste your original pair of images.
Paste 3 different views of the 3D-reconstructed scene
• Does the 3D reconstruction look right to you? • Explain how the different elements of the scene are correctly reconstructed in 3D • Explain any issues that you have found to reconstruct the scene in 3D

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Paste the reprojection error histogram for your custom example
• Interpret the results of the reprojection error histogram above • Do you obtain reasonable error values? • Compare the results with the ones obtained for the provided example.

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AI usage reflection (Specific answers to this Lab)

We want you to use AI responsibly and reflect about its usage in an **honest** manner.

• Which AI tools have you used? (ChatGPT / Copilot / Gemini, embedded in Visual Studio, Cursor, etc) • For which tasks have you used these tools? (Debugging, writing code, understanding concepts)
• What did the tool succeed on (in this Lab)? • What did the tool fail in (in this Lab)??
• Have you changed how you use these tools compared to the previous lab? Explain it.

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Provide s**creenshots of the prompts** you used and the resulting output from the tool:

- **Successful Example**: One instance where the tool significantly assisted you. Justify its success and usefulness.

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- **Failed Example**: One instance where the tool did not perform as expected. Explain the reason for the failure.

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Self-awareness

- Provide an overall explanation of the lab. Describe the goals of the lab clearly. State the problem you were trying to solve or explore in this lab.

- What have you learned? Did the lab help you visualise and understand concepts?

- Which parts of the notebook did you succeed in? Describe the sections where you felt confident and explain why you think they were successful.

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• Which parts of the notebook did you fail to solve? Be honest about the areas where you faced difficulties. What challenges or issues did you encounter that you couldn't resolve? How would you approach these issues in the future?
• Are there any concepts that are not completely clear yet? I.e., what question would you least like to be asked about?

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Extras

Add extra results below. If needed, you can use multiple pages for this section.